

Solving Problems with Equations

MA.7.AR.2.2

I can distribute, choose a starting method, and write an equation from a story.

Yesterday's last whiteboard question had a bracket in it.

Today the bracket is the lesson.

Warm-Up

On your own. Four minutes.

Same and different

What is the same about these two, and what is different?

$$5(2y - 7)$$

$$5 \cdot 2y - 7$$

Notes

I. The factor outside multiplies **EVERY** term inside

A. **distributive property** — $a(b + c) = a \cdot b + a \cdot c$

B. **Every term inside. Not just the first one.**

C. The factor may sit on either side: $a(b + c)$ and $(b + c)a$ are the same.

Notes

II. A negative outside multiplies both terms too

- A. $-4(x - 9) = -4x + 36$
- B. The second term was already negative, and a negative times a negative is positive.
- C. **The sign that changes is the one you were not watching.**

Notes

III. Two ways to start $a(b + c) = d$

- A. **Method 1. Distribute the a , then solve as usual.**
- B. **Method 2. Divide both sides by a , then solve as usual.**
- C. Both give the same answer. Neither is required.
- D. **Choose the one that keeps the numbers friendly.**

Notes

IV. Parentheses appear when a whole group repeats

- A. **Find the thing that changes. Give it a letter.**
- B. Find what stays the same no matter what. That is a constant.
- C. **If a whole group is repeated, the group goes in brackets and the count goes outside.**
- D. Write the total last, on the other side of the equal sign.

Example 1

Which method would you start with?

$$6(\mathbf{x} - 2) = 42$$

Method 1 or Method 2, and why? Answer out loud.

Example 1

Both methods, side by side.

Method 1: distribute	Method 2: divide
$6x - 12 = 42$	$x - 2 = 7$
$6x = 54$	$x = 9$
$x = 9$	

Example 1

Check it.

$$6(9 - 2) = 6(7) = 42 \quad \checkmark$$

Example 2

The negative outside.

$$-3(\mathbf{x} + 4) = 18$$

Distribute it. Whiteboards, first line only.

Example 2

Both terms.

$$-3(x + 4) = 18$$

$$-3x - 12 = 18$$

both terms, and both signs

$$-3x = 30$$

$$x = -10$$

Example 2

Check it.

$$-3(-10 + 4) = -3(-6) = 18 \quad \checkmark$$

Example 3

Six tickets, each \$1.75 off.

Six tickets are bought. Each one has \$1.75 taken off its price.

The total paid is \$46.92. Write the equation. Do not solve it.

Example 3

Inside or outside?

$6(p - 1.75) = 46.92$	$6p - 1.75 = 46.92$
\$1.75 off EACH ticket	\$1.75 off the whole order, once

Example 3

Solve it.

$$6(p - 1.75) = 46.92$$

$$p - 1.75 = 7.82 \quad \text{Method 2: divide by 6 first}$$

$$p = 9.57$$

Each ticket costs \$9.57 before the discount.

Whiteboards • Question 1 of 9

Boards up on three.

$$6(x - 2) = 42$$

Solve it.

One value. Say which method you started with.

Whiteboards • Question 1 of 9

Answer.

$$6(x - 2) = 42$$

divide by 6: $x - 2 = 7$

Answer: $x = 9$

Whiteboards • Question 2 of 9

Boards up on three.

$$-6(x - 2) = 42$$

Solve it.

One value. Say which method you started with.

Whiteboards • Question 2 of 9

Answer.

$$-6(x - 2) = 42$$

divide by -6 : $x - 2 = -7$

Answer: $x = -5$

Whiteboards • Question 3 of 9

Boards up on three.

$$6(x + 2) = 42$$

Solve it.

One value. Say which method you started with.

Whiteboards · Question 3 of 9

Answer.

$$6(x + 2) = 42$$

divide by 6: $x + 2 = 7$

Answer: $x = 5$

Whiteboards • Question 4 of 9

Boards up on three.

$$0.6(x - 2) = 4.2$$

Solve it.

One value. Say which method you started with.

Whiteboards • Question 4 of 9

Answer.

$$0.6(x - 2) = 4.2$$

divide by 0.6: $x - 2 = 7$

Answer: $x = 9$

Whiteboards • Question 5 of 9

Boards up on three.

$$(3/5)(x - 2) = 4.2$$

Solve it.

One value. Say which method you started with.

Whiteboards • Question 5 of 9

Answer.

$$(3/5)(x - 2) = 4.2$$

3/5 IS 0.6

Answer: $x = 9$

Whiteboards • Question 6 of 9

Boards up on three.

$$6(x - 2) = 43$$

Solve it.

One value. Say which method you started with.

Whiteboards • Question 6 of 9

Answer.

$$6(x - 2) = 43$$

divide by 6: $x - 2 = 43/6$

Answer: $x = 9 \frac{1}{6}$

Whiteboards • Question 7 of 9

Boards up on three.

$$4(1 - 3x) = 52$$

Solve it.

One value. Say which method you started with.

Whiteboards • Question 7 of 9

Answer.

$$4(1 - 3x) = 52$$

divide by 4: $1 - 3x = 13$, so $-3x = 12$

Answer: $x = -4$

Whiteboards • Question 8 of 9

Boards up on three.

$$0.2(r - 3) = -10$$

Solve it.

One value. Say which method you started with.

Whiteboards · Question 8 of 9

Answer.

$$0.2(r - 3) = -10$$

divide by 0.2: $r - 3 = -50$

Answer: $r = -47$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

A plumber charges \$75 for the visit plus \$66.24 for every hour worked. The bill was \$339.96. How many hours?

Write your answer in sentences.

This one is written work. Write the equation first, then solve it.

Whiteboards • Question 9 of 9

Answer.

A plumber charges \$75 for the visit plus \$66.24 for every hour worked. The bill was \$339.96. How many hours?

$$75 + 66.24h = 339.96, \text{ so } 66.24h = 264.96$$

Answer: $h = 4$ hours

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.